Algorithm 1 SDF-Guided Sampling MPC for Human–Object Retargeting

Require: reference $x^{\text{ref}}_{0:T}$ (qpos, object pose, contact mask m), horizon H, samples N_W, iters N, elite frac ρ, min-valid frac f, temperature λ, anneal β_1, β_2, tol ϵ

initialise control knots $U \leftarrow \text{interp}^{-1}(u^{\text{ref}})$
for each simulation step $s = 0, 1, \ldots$ **do** ▷ outer receding horizon
  $J_{\text{best}} \leftarrow \infty,\ i \leftarrow 0$
  while $i < N$ **and** not converged **do** ▷ inner CEM
    $i \leftarrow i + 1$; update Σ^i_h by anneal schedule (β_1, β_2)
    for $j = 1, \ldots, N_W$ **do**
      $[W]_j \sim \mathcal{N}(0, \Sigma^i)$; $[U^i]_j \leftarrow U^i + [W]_j$
      roll out $x^j_{0:H}$ in MuJoCo-Warp from $u^j = \text{interp}([U^i]_j, \tau)$
      $J^j,\ \mathbf{g}^j \leftarrow \textsc{CostAndGate}(x^j_{0:H}, x^{\text{ref}})$ ▷ Alg. 2
    end for
    $U^{i+1} \leftarrow \textsc{GatedElite}(\{J^j\}, \{\mathbf{g}^j\}, \{[U^i]_j\}, \rho, f, \lambda)$ ▷ Alg. 2
    if $|J_{\text{best}} - \min_j J^j| < \epsilon$ **then**
      break ▷ early stop
    end if
    $J_{\text{best}} \leftarrow \min_j J^j$
  end while
  execute first n_{exec} controls of U; advance simulator
end for
return refined dynamically-feasible controls U^*

Algorithm 2 Cost with Multi-Scale Contact Reward, and Gated-Elite Selection

function CostAndGate($x_{0:H}, x^{\text{ref}}$)
    $J \leftarrow 0$
    for $t = 0, \ldots, H$ **do**
        $J \mathrel{+}= \underbrace{\left\|(q_t - q_t^{\text{ref}})\, W_{\text{loc}}\right\|}_{\text{local-frame tracking}} + \underbrace{\alpha_o \,\|\, o_t - o_t^{\text{ref}}\|}_{\text{object pose}}$
        for each end-effector e with contact $m_t^e = 1$ **do**   ▷ COARSE: HDMI anchor
            $r^{\text{c}}_{t,e} \leftarrow g\, \exp\big(-\|p_t^e - p_t^{\star,e}\|/\sigma_c\big)$ (+ palm-orientation term)
            $\phi \leftarrow \phi_e(x_t)$ ▷ FINE: object-local grid-SDF
            $r^{\text{f}}_{t,e} \leftarrow s\, \exp\big(-|\phi|/\sigma_f\big) \cdot \mathbf{1}[\phi \in B]$
            $J \mathrel{-}= r^{\text{c}}_{t,e} + r^{\text{f}}_{t,e}$
        end for
        for each part group $P \in \{\text{hand}, \text{body}, \text{leg}\}$ **do**   ▷ part-wise constraint, soft layer
            $J \mathrel{+}= w_P\, \max\big(0,\ \delta_P - \phi_P(x_t)\big)$
        end for
        $J \mathrel{+}=$ E167A body-z / ground-z penetration terms
    end for
    for each part group P **do**   ▷ same constraint, hard layer: gate stats
        $g_P^{\min} \leftarrow \min_t \phi_P(x_t)$; $g_P^{\text{vio}} \leftarrow \frac{1}{H}\sum_t \mathbf{1}[\phi_P(x_t) < \theta_P]$
    end for
    return $J,\ \mathbf{g} = \{(g_P^{\min}, g_P^{\text{vio}})\}_P$
end function

function GatedElite($\{J^j\}, \{\mathbf{g}^j\}, \{U_j\}, \rho, f, \lambda$)
    $\text{valid}_j \leftarrow \bigwedge_P \left[\, g_P^{\min,j} \geq \text{floor}_P \ \wedge\ g_P^{\text{vio},j} \leq \tau_P \,\right]$ ▷ per-part AND
    if $\#\{j : \text{valid}_j\} \geq f\, N_W$ **then**
        $\mathcal{E} \leftarrow$ top-$\lceil \rho N_W \rceil$ of $\{j : \text{valid}_j\}$ by lowest J^j
    else   ▷ least-violation fallback
        $\mathcal{E} \leftarrow$ top-$\lceil \rho N_W \rceil$ by lowest $\big(\text{depth}^j + \text{vio}^j - 10^{-3}\, \tilde{J}^j\big)$
    end if
    $w_j \leftarrow \dfrac{\exp(-J^j/\lambda)\, \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} \exp(-J^k/\lambda)}$
    return $\sum_j w_j\, U_j$ ▷ softmax elite refit
end function